

A cubic graph on sixteen vertices without an equitable four-total coloring

Xiaojian Zeng

PBC School of Finance, Tsinghua University, 100190

xiaojianzeng@foxmail.com

Abstract

We give a connected simple cubic graph on sixteen vertices whose total chromatic number is four but which has no equitable four-total coloring. This answers negatively Question 7.1 of Adaauto, de Figueiredo, Sasaki and Schneider (arXiv:2609.05259v1). An explicit coloring proves the upper bound of four. To exclude equitable colorings, two pairs of vertices with identical neighborhoods exhaust a vertex color class. The resulting three-total coloring of a six-cycle gives equal colors to the endpoints of a chord, a contradiction. The proof is self-contained; an independently checked Boolean unsatisfiability certificate provides a computational cross-check. We do not determine the minimum counterexample order or the exact equitable total chromatic number of the displayed graph.

1 The question and the counterexample

All graphs in this note are finite, simple and undirected. A proper k -total coloring assigns colors from a palette of size k to $V(G) \cup E(G)$ so that adjacent vertices, adjacent edges, and an edge and either of its endpoints have different colors. It is *equitable* if the sizes of any two color classes, counting vertices and edges together, differ by at most one. The total chromatic number $\chi''(G)$ is the least k admitting a proper k -total coloring. A cubic graph is *Type 1* when $\chi''(G) = 4$.

Adaauto, de Figueiredo, Sasaki and Schneider [1] ask in Question 7.1 whether every Type 1 cubic graph of order less than 20 admits at least one equitable 4-total coloring. Their examples $L_{12}, L_{18}, H_{16}, H_{18}$ admit non-equitable colorings, but also admit equitable colorings, as stated in their Section 7. In particular, their Theorem 5.3 describes possible color-class profiles at order 16 and gives H_{16} realizing both profiles; it does not assert that every Type 1 graph of that order realizes the equitable profile. The following graph excludes that profile altogether.

Theorem 1. *Let G have vertex set $\{0, 1, \dots, 15\}$ and the 24 edges in Table 1. Then G is connected and cubic, $\chi''(G) = 4$, and G has no equitable 4-total coloring.*

The graph is also specified by the graph6 string `0s?GOCF@0??ACBCBL?B?0`. Table 1 gives a proper total coloring f with palette $\{0, 1, 2, 3\}$. This graph, rather than any enumeration of smaller graphs, will establish the theorem.

The edge list shows that G is simple and every vertex has degree three. For connectedness, the vertices 1, 2 share neighbors 0, 14, 15, and 12, 13 share neighbors 3, 10, 11. The edges $\{0, 3\}$ and $\{10, 15\}$ join these two subgraphs. The remaining vertices form the cycle

4, 5, 9, 7, 8, 6, 4,

Table 1: The complete edge set of G and a proper total coloring f . Each unordered pair in the upper part is an edge; the lower part gives all vertex colors.

Edge e	$f(e)$	Edge e	$f(e)$	Edge e	$f(e)$
$\{0, 1\}$	1	$\{3, 13\}$	2	$\{7, 8\}$	1
$\{0, 2\}$	2	$\{4, 5\}$	1	$\{7, 9\}$	2
$\{0, 3\}$	3	$\{4, 6\}$	2	$\{9, 11\}$	3
$\{1, 14\}$	0	$\{4, 14\}$	3	$\{10, 12\}$	2
$\{1, 15\}$	2	$\{5, 8\}$	2	$\{10, 13\}$	1
$\{2, 14\}$	1	$\{5, 9\}$	0	$\{10, 15\}$	3
$\{2, 15\}$	0	$\{6, 7\}$	0	$\{11, 12\}$	1
$\{3, 12\}$	0	$\{6, 8\}$	3	$\{11, 13\}$	0

v	0	1	2	3	4	5	6	7
$f(v)$	0	3	3	1	0	3	1	3

v	8	9	10	11	12	13	14	15
$f(v)$	0	1	0	2	3	3	2	1

which attaches through $\{4, 14\}$ and $\{9, 11\}$. Thus all vertices belong to one component. In the displayed coloring, endpoints of each edge differ, and each vertex together with its three incident edges has all four colors. These checks verify every total-coloring constraint. The total color-class sizes, in color order, are $(10, 10, 9, 11)$; the vertex color-class sizes are $(4, 4, 2, 6)$. A vertex and its three incident edges require four distinct colors, so this witness proves $\chi''(G) = 4$. The remainder of the proof excludes every equitable coloring of G .

2 A counting identity and a twin constraint

Lemma 2. *In a proper 4-total coloring of a cubic graph of order n , let v_a and e_a be the numbers of vertices and edges of color a , and put $t_a = v_a + e_a$. Then*

$$2e_a = n - v_a, \quad 2t_a = n + v_a. \quad (1)$$

For $n = 16$, the coloring is equitable if and only if $v_a = 4$ for every color a .

Proof. The vertex and its three incident edges use all four colors. Hence every vertex not colored a meets exactly one edge of color a , whereas a vertex colored a meets none. Counting endpoints of the edges colored a proves the first identity, and the second follows by adding $2v_a$. At order 16 there are $16 + 24 = 40$ elements to color. Four integer class sizes summing to 40 and differing by at most one must all be 10. The second identity therefore gives the stated equivalence. $\square$

The identity (1) is also contained in [1, Lemma 3.1]; the argument above records exactly the form needed here.

Lemma 3. *Suppose two nonadjacent vertices u, v of a simple cubic graph have the same three neighbors. In every proper 4-total coloring, u and v have the same color a . At each common neighbor, the edge other than the edges to u and v has color a .*

Proof. Suppose instead that u and v have different colors a, b , and call the remaining colors c, d . Each common neighbor has color c or d . There is a c -colored edge at each of u and v . Their other endpoints must have color d and must be distinct, since two incident edges cannot share a color.

Similarly, the d -colored edges at u and v need two distinct common neighbors of color c . This requires at least four common neighbors, a contradiction.

Now let both u, v have color a , and let a common neighbor have color b . Its edges to u, v have distinct colors and avoid a, b , so they use the other two colors. Its third edge must consequently have color a . $\square$

3 Excluding an equitable coloring

Completion of the proof of Theorem 1. Suppose that G has an equitable 4-total coloring. Apply Lemma 3 to 1, 2, whose common neighbors are 0, 14, 15. For some color a , vertices 1, 2 and edges

$$\{0, 3\}, \quad \{4, 14\}, \quad \{10, 15\}$$

all have color a . Apply the lemma also to 12, 13, whose common neighbors are 3, 10, 11. The edge $\{0, 3\}$ is a third edge in both applications, so the second pair has the same color a ; the edge $\{9, 11\}$ also has color a .

By Lemma 2, the four vertices 1, 2, 12, 13 exhaust the vertex class of color a . Thus every vertex in $S = \{4, 5, 6, 7, 8, 9\}$ avoids a and must meet exactly one edge of color a . Vertices 4 and 9 already meet such edges outside S , namely $\{4, 14\}$ and $\{9, 11\}$. All neighbors of 5, 6, 7, 8 lie in S . Consequently the a -colored edges inside S form a perfect matching on $\{5, 6, 7, 8\}$. In the subgraph induced by these four vertices, vertex 5 has only neighbor 8. The matching is therefore necessarily

$$\{\{5, 8\}, \{6, 7\}\}.$$

It follows that no vertex or edge of the six-cycle

$$4, 5, 9, 7, 8, 6, 4 \tag{2}$$

has color a . Restricting the coloring to this cycle gives a proper total coloring using the other three colors. In any such coloring, every three consecutive vertices have distinct colors. Indeed, adjacent vertices already differ; if the first and third had the same color, both edges at the middle vertex would be forced to use the same remaining color. With three colors, the vertex colors therefore repeat with period three. Vertices 5 and 8, three steps apart in (2), must have the same color. Since $\{5, 8\}$ is an edge of G , this contradicts properness. No equitable 4-total coloring exists. $\square$

Since $16 < 20$, Theorem 1 answers the stated universal question negatively, including when the graph universe is restricted to connected graphs. The argument makes no assertion that 16 is the minimum counterexample order. It also does not determine the exact equitable total chromatic number of G , or classify the graphs of orders 16 or 18.

4 Certificates and computational assistance

The graph was found in a computer search using **geng** from nauty 2.9.3 and total-coloring tests. The single-graph proof above does not require completeness of that search. Machine-readable graph and color data, a Boolean formula, a DRAT proof, and verification code are included in the accompanying reproducibility files.

For clarity, the equitable-coloring formula can be specified completely. For each of the 40 elements $z \in V(G) \cup E(G)$ and each color a , introduce a Boolean variable $x_{z,a}$. Require exactly one

color per element, and forbid a shared color for each conflicting pair. Finally, for each color a and each five-element subset $U \subseteq V(G)$, include the clause

$$\bigvee_{v \in U} \neg x_{v,a}.$$

These last clauses force at most four vertices of each color and thus exactly four, which is equivalent to equitability by Lemma 2. There are 160 variables and 18 232 clauses: 280 exactly-one clauses, 480 conflict clauses, and $4\binom{16}{5} = 17\,472$ vertex-balance clauses. No symmetry-breaking clauses are imposed. CaDiCaL 3.0.1 produced an unsatisfiability proof, which `drat-trim` verified. A separate verification implementation reconstructed the entire clause set, checked the displayed coloring, and replayed the DRAT check. This corroborates the structural proof.

For adjacent work on certified Boolean search in finite combinatorics, see Zeng [2]. That reference concerns Erdős–Szekeres configurations; the connection here is the use of checkable SAT certificates. The comparison with prior total-coloring work is specifically to Question 7.1 and the stated examples of [1]. This bounded comparison does not establish exhaustive literature priority.

AI assistance. OpenAI Codex assisted with the computational search, proof development, verification code, and manuscript preparation. The argument and certificates also received a separate AI-assisted verification pass. That pass is not external peer review. The mathematical proof is given in full above, and the computational checks have the scope described in this section.

References

- [1] M. Adauto, C. de Figueiredo, D. Sasaki and R. Schneider, *Order 14 is the largest order for which every 4-total coloring of every cubic graph is equitable*, arXiv:2609.05259v1, 2026. <https://arxiv.org/abs/2609.05259v1>.
- [2] Z. Zeng, *One-Sided Parity Compression for Certified SAT Attacks on the 33-Point Erdős–Szekeres Problem*, 2026. <https://doi.org/10.2139/ssrn.7334758>.